

SE200 CFW11 Industrial VFD
Version 1.0
Operations Manual

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Keypad (HMI)

The main screen of the SE200 CFW11's HMI includes several buttons and indicators to provide feedback and control of the VFD. An example of the main screen is shown in the figure below.

The integral keypad can be used to operate and program (view/edit all parameters) of the CFW-11 inverter.

The inverter keypad navigation is similar to the one used in cell phones and the parameters can be accessed in numerical order or through groups (Menu).

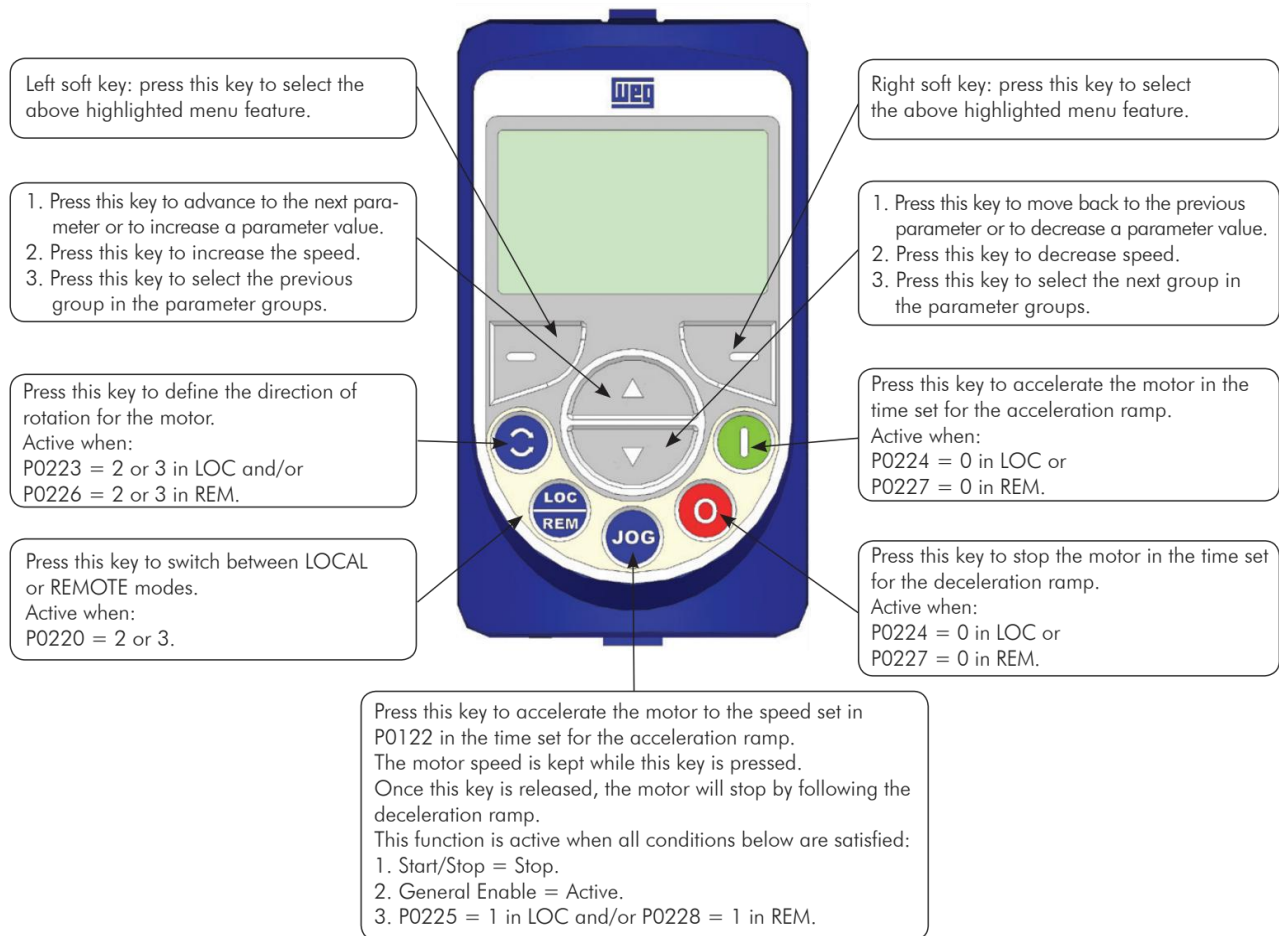


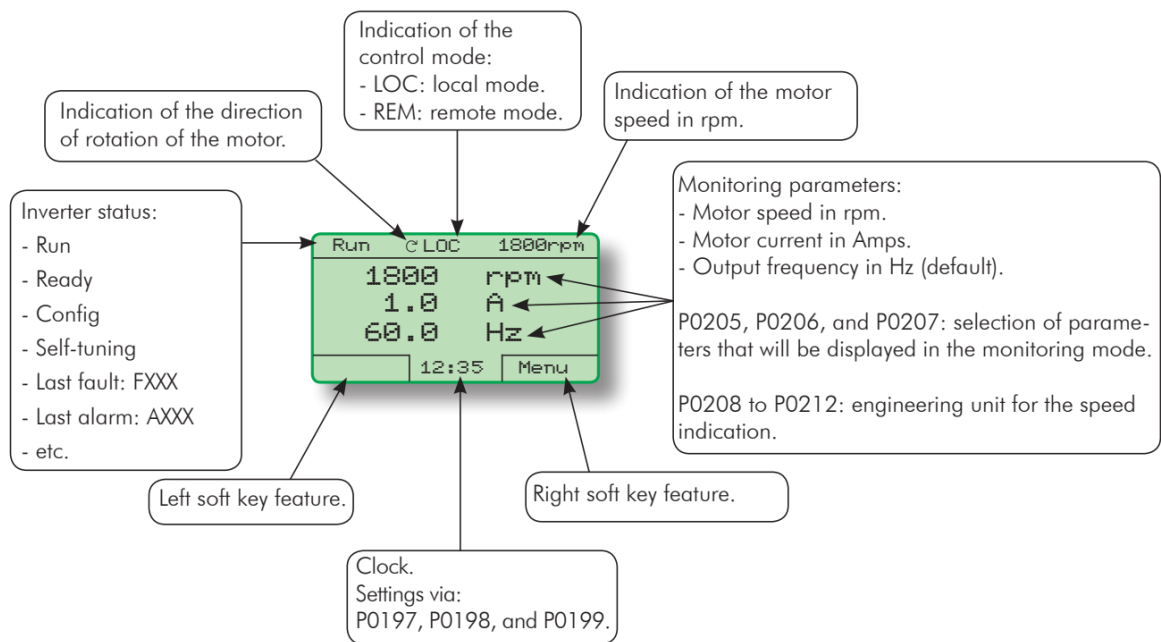
Figure 4.1 - Operator keys

Battery:

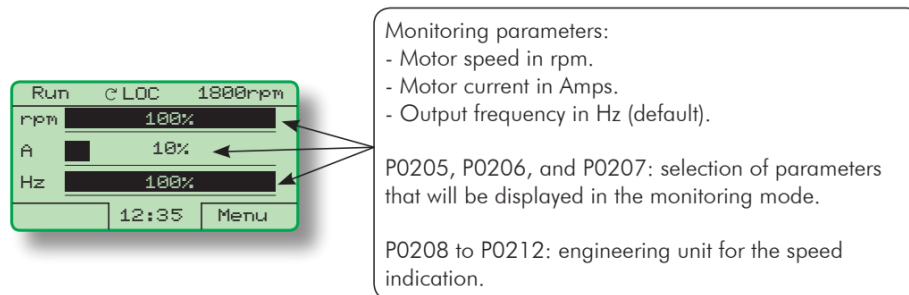


NOTE!

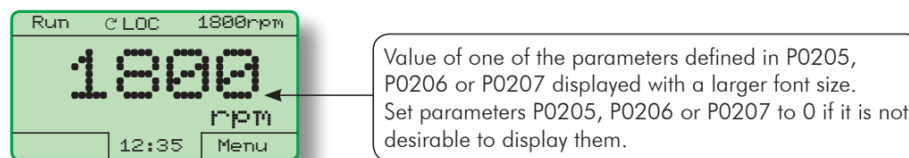
The battery is necessary only to keep the internal clock operation when the inverter stays without power. If the battery is completely discharged or if it is not installed in the keypad, the displayed clock time will be invalid and an alarm condition "A181 - Invalid clock time" will be indicated whenever the AC power is applied to the inverter.



(a) Monitoring screen with the factory default settings



(b) Example of a monitoring screen with bar graphs



(c) Example of a monitoring screen displaying a parameter with a larger font size

Figure 1/2 Keypad

Power and Motor Terminations

The SE200 CFW11 VFD supports both single-phase up to frame size E and three-phase applications. When using it in a single-phase setup, follow these key guidelines to ensure proper operation:

1. Power Input Connections

- Connect single-phase power to A/R/L1 and B/S/L2 terminals, these terminals supply power to the DC bus of the CFW11.
- Three phase applications just use R/S/T respectively

2. Motor Wiring

- Terminate the motor leads at the U, V, and W terminals, located at the bottom right of the VFD.

3. Sizing Requirement

- When using the SE200 CFW11 in single-phase mode, select a VFD rated at twice the intended motor load.
- This ensures sufficient capacity and reliable performance.

4. Parameter Configuration

- Set P0357 = 0 sec to bypass the phase loss fault (this is required for operation).

Following these steps will help ensure efficient and reliable operation of the SE200 CFW500 VFD in a single-phase configuration

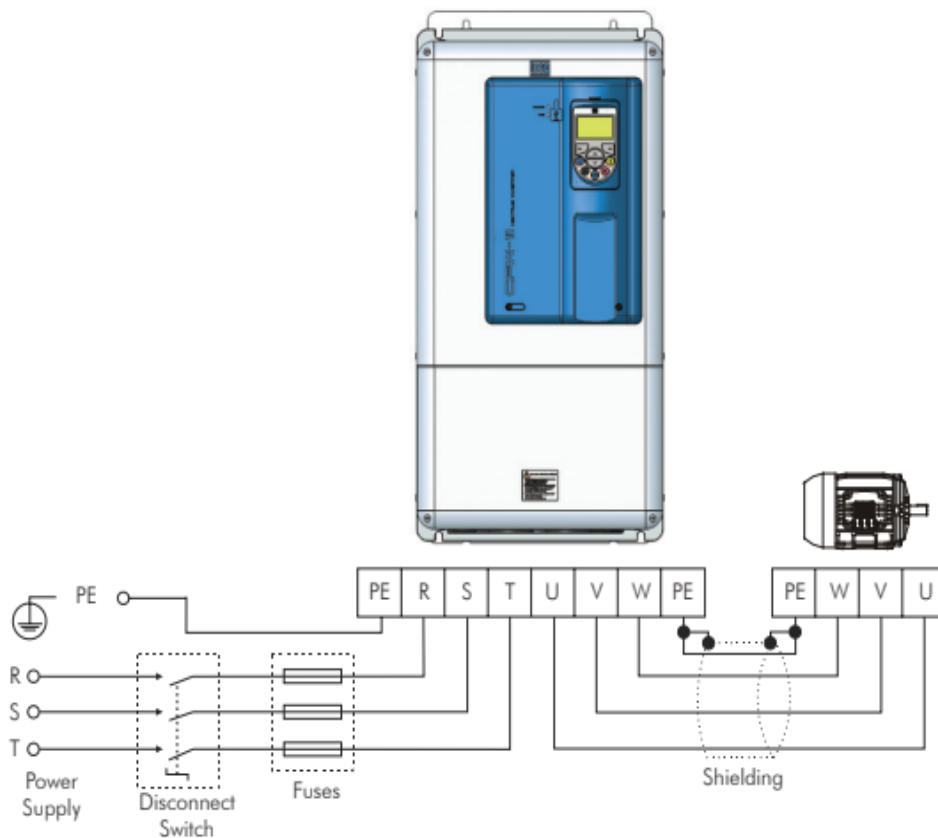


Figure 2 Power and Motor Connections

Initial Power Up

When powering up the SE200 CFW11 for the first time, the field technician must configure the device using the **Field Technician Parameters** section outlined in this manual.

Steps to Access and Configure Parameters:

1. Access the Parameter Section

- Press **Menu > Orientated Startup**.
- Use the **Up** and **Down** arrows to navigate to the bottom to **Startup**.
- Set **P0317** to 1 to enter the orientated **Startup**.

2. Complete the Oriented Startup Menu

- Starting at Step 26, the SE200 CFW11 Industrial VFD will guide you through the oriented startup steps (Steps 26–33) highlighted in blue in this guide.
- It's important to note that we recommend most industrial applications be controlled in V/F mode (P0202 = 0) however in some cases, when there is nuisance dc bus over voltage tripping the unit may need to be operated in Sensorless mode (P0202 = 3).
- When done press the **Back/Esc** button to exit the menu back to the main menu

3. Continue Programming

- Return to **Menu > Parameters** to proceed with the programming procedure (Steps 1–25). If you have a completed package the steps in **white** will already be completed. Only the **blue** (Steps 26-35) will be required.

PARAMETERS TO BE SET BY FIELD PERSONNEL (Steps #1-42)					
Step#	Parameter #	Parameter Description	Unit	Default Value	Set Value
1	P0000	Access to Parameters	0-9999	0000	0005
2	P0100	Acceleration Time	Sec	20.0	Set to desired time
3	P0101	Deceleration Time	Sec	20.0	Set to desired time

4	P0133	Minimum Speed	0-18000 rpm	90	Set to desired min speed in rpm
5	P0134	Maximum Speed	0-18000 rpm	1800	Set to desired min speed in rpm
6	P0217	Zero Speed Disable	Disable Magnetization Current	0	1
7	P0220	LOC/REM Selection Source	LR Key LOC	2	2
8	P0221	LOC Reference Select	AI1	0	1
9	P0222	Remote Reference Select	AI1	0	1
10	P0223	LOC FWD/REV Selection	Always FWD (change if required)	2	0
11	P0224	LOC Run/Stop Selection	DIX	0	1
12	P0225	LOC JOG Selection	Disable	1	0
13	P0226	REM FWD/REV Selection	Always FWD (change if required)	4	0
14	P0227	REM Run/Stop Selection	DIX	1	1
15	P0228	REM JOG Selection	Disable	2	0

16	P0229	Stop Mode Selection	Coast to Stop (change if required)	0	1
17	P0233	AI1 Signal Type	Set to either 0-10vdc for speed pot or 4-20mA for plc	0	0
18	P0238	AI2 Signal Type	Set to either 0-10vdc for speed pot or 4-20mA for plc	0	1
19	P0263	DI1 Function	Run/Stop	1	1
20	P0353	IGBT's / Air Overtemp Config	Under Temp Disable	0	4
21	P0317	Oriented Start Up	Yes	0	1
22	P0201	Language	English	1	1
23	P0202	Type of Control	V/f 60Hz	0	0
24	P0296	Line Rated Voltage	0-6 (Can be adjusted if overvoltage or undervoltage trips occur)	Depending on inverter model	Set as per input voltage
25	P0298	Application	Normal Duty	0	0
26	P0398	Motor Service Factor	As per motor nameplate		Motor SF
27	P0400	Motor Rated Voltage	As per motor nameplate		Motor Voltage
28	P0401	Motor Rated Current	As per motor nameplate		Motor Current
29	P0402	Motor Rated Speed	As per motor nameplate		Motor RPM

30	P0403	Motor Rated Frequency	As per motor nameplate		Motor Hz
31	P0404	Motor Rated Power	As per motor nameplate		Motor HP/KW
32	P0406	Motor Ventilation	Self-Vent	0	0
33	P0407	Rated Power Factor	0.5 to 0.99	0.68	Set as per nameplate
34	P0156	Overload Current 100% Speed	0.1 to 1.5 x Inom-ND Example: Motor FLA – 13.2A If SF is 1.0, P0156 should be set to 15.2A If SF is 1.15, P0156 should be set to 16.5A	1.05 x Inom-ND	As per motor service factor, adjust if required (SF=1.0 – 1.15 x Inom-ND) (SF=1.15 – 1.25 x Inom-ND)
35	P0159	Motor Thermal Class	Class 5,10,15,20,25,30,35,40,45	1 – Class 10	3 – Class 20

Table 1 Basic SE200 Industrial Startup guide

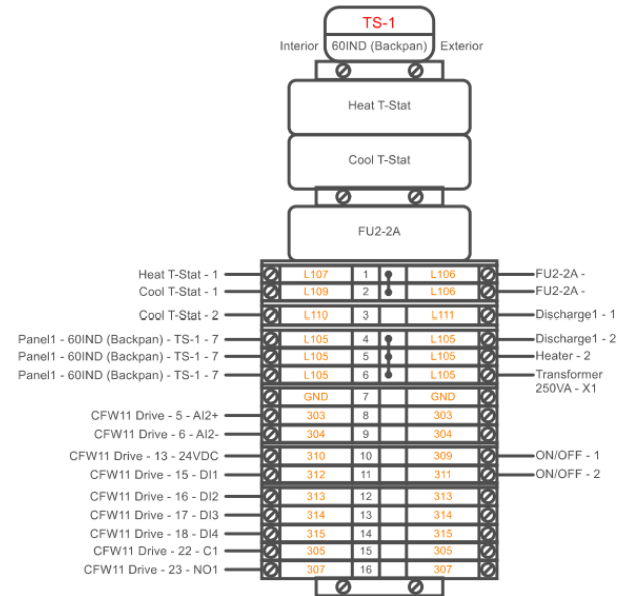
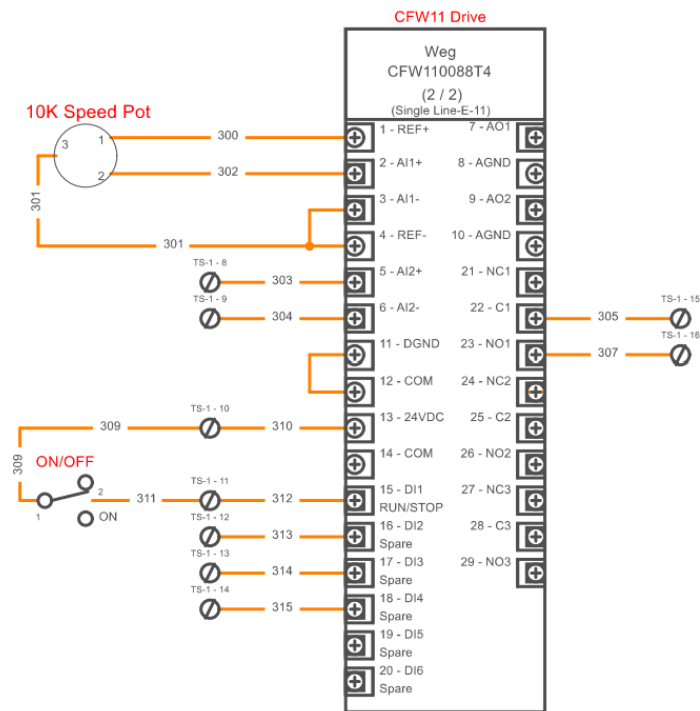


Figure 3 Industrial Standard Terminal Layout

PID Control

1. Enable PID:

- To select the especial function: PID Regulator (P0203 = 1)
- When the PID function is enabled, by setting P0203 = 1 via HMI, the following parameters are changed automatically: P0205 = 10 (Reading Parameter Selection 1: Setpoint PID #).
- P0206 = 9 (Reading Parameter Selection 2: Process Var. #).
- P0207 = 2 (Reading Parameter Selection 3: Motor Speed #).
- P0223 = 0 (FORWARD/REVERSE Selection - LOCAL Situation: Always FORWARD). P0225 = 0 (JOG Selection – LOCAL Situation: Disabled).
- P0226 = 0 (FORWARD/REVERSE Selection - REMOTE Situation: Always FORWARD). P0228 = 0 (JOG Selection - REMOTE Situation: Disabled).
- P0236 = 3 (AI2 Signal Function: Process Variable).
- P0265 = 22 (DI3 Function: Manual/Automatic)

For the operation of the PID Controller application, it is necessary to set the parameter P0203 = 1.

The DI3 function, defined by the parameter P0265, works in the following manner:

DI3	Operation
0 (0 V)	Manual
1 (24 V)	Automatic

2. Define the PID action type

To define the type of PID action that the process requires: direct or reverse. The control action must be direct (P0527 = 0) when it is necessary that the speed increases for the process variable to increase. Otherwise, select reverse (P0527 = 1).

Examples:

a) Direct: A pump driven by an inverter filling up a reservoir, with the PID controlling its level. In order that the level (process variable) increases, it is necessary that the flow increases, which is obtained with the increase of the motor speed.

b) Reverse: A fan driven by an inverter doing the refrigeration of a cooling tower, with the PID controlling its temperature. In order that the temperature (process variable) increases, it is necessary that the ventilation be reduced, by means of the reduction of the motor speed.

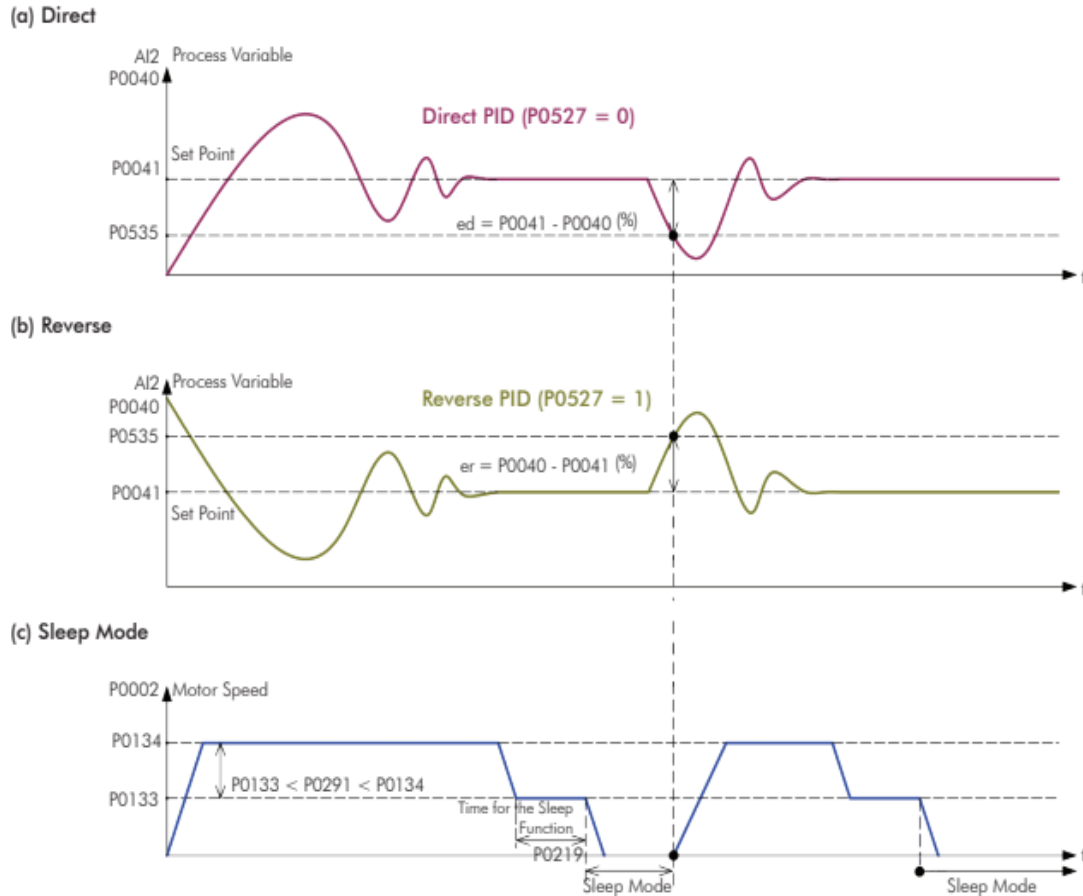


Figure 4 Action types

3. Define Feedback

To define the feedback input: the feedback (process variable measurement) is always done via one of the analog inputs (selected at P0524). To make this guide easier, the AI2 input will be selected (P0524 = 1).

4. Define speed limits

To set the speed limits: Adjust P0133 and P0134 according to the application.

The readings exhibited automatically when the inverter is powered are:

- Reading 1 – P0041 "PID Setpoint".
- Reading 2 – P0040 "Process Variable".
- Reading 3 – P0002 "Motor Speed".

5. Typical SEEL PID parameter setup for CFW11

4. Access the Parameter Section

- Press **Menu > Orientated Startup**.
- Use the **Up** and **Down** arrows to navigate to the bottom to **Startup**.
- Set **P0317** to 1 to enter the orientated **Startup**.

5. Complete the Oriented Startup Menu

- Starting at Step 26, the SE200 CFW11 Industrial VFD will guide you through the oriented startup steps (Steps 26–33) highlighted in blue in this guide.
- It's important to note that we recommend most industrial applications be controlled in V/F mode (P0202 = 0) however in some cases, when there is nuisance dc bus over voltage tripping the unit may need to be operated in Sensorless mode (P0202 = 3).
- When done press the **Back/Esc** button to exit the menu back to the main menu

6. Continue Programming

Return to **Menu > Parameters** to proceed with the programming procedure (Steps 1–36). If you have a completed package the steps in **white** will already be completed. Only the **blue** (Steps 37-51) will be required.

<u>PARAMETERS SET AT SEEL FACILITY (#1-36)</u>					
Step#	Parameter #	Parameter Description	Unit	Default Value	Set Value
1	P0000	Access to Parameters	0-9999	0000	0005
2	P0100	Acceleration Time	Sec	20.0	Set to desired time

3	P0101	Deceleration Time	Sec	20.0	Set to desired time
4	P0133	Minimum Speed	0-18000 rpm	90	Set to desired min speed in rpm
5	P0134	Maximum Speed	0-18000 rpm	1800	Set to desired min speed in rpm
6	P0203	Special Function Select	PID Regulator	0	1
7	P0217	Zero Speed Disable	Disable Magnetization Current	0	1
8	P0220	LOC/REM Selection Source	Dlx	2	4
9	P0221	LOC Reference Select	Keypad or AI1 if speed pot installed	0	0 or 1
10	P0222	Remote Reference Select	Keypad	1	0
11	P0223	LOC FWD/REV Selection	Always FWD (change if required)	2	0
12	P0224	LOC Run/Stop Selection	DIX	0	1
13	P0225	LOC JOG Selection	Disable	1	0
14	P0226	REM FWD/REV Selection	Always FWD (change if required)	4	0

15	P0227	REM Run/Stop Selection	DIX	1	1
16	P0228	REM JOG Selection	Disable	2	0
17	P0229	Stop Mode Selection	Coast to Stop (change if required)	0	1
18	P0233	AI1 Signal Type	Set to either 0-10vdc or 4-20mA	0	0
19	P0236	AI2 Signal Function	Process Variable	0	3
20	P0238	AI2 Signal Type	Set to either 0-10vdc or 4-20mA	0	1
21	P0263	DI1 Function	Run/Stop	1	1
22	P0265	DI3 Function	LOC/REM	0	9
23	P0266	DI4 Function	MAN/AUTO	0	22
24	P0268	DI6 Function	General Enable	14	2
25	P0353	IGBT's / Air Overtemp Config	Under Temp Disable	0	4
26	P0320	Fly Start/Ride-Through	Ride-Through	0	3
27	P0524	PID Feedback Select	AI2	1	1
28	P0525	Keypad PID Setpoint	0-100%	0	0-100% Set to desired PID setpoint for operation

29	P0527	PID Action Type	Direct/Reverse (Direct speeds up when value is lower than setpoint, Reverse speeds up when higher than setpoint)	0	0 or 1
30	P0528	Process Variable Scale Factor	1 to 9999	1000	1 to 9999
31	P0529	Process Variable Decimal Point	0-4	1	0-4
32	P0530	Process Variable Eng Unit 1	32-127	37	32-127
33	P0531	Process Variable Eng Unit 2	32-127	37	32-127
35	P0532	Process Variable Eng Unit 3	32-127	37	32-127
36	P0536	P0525 Automatic Setting	Off/On	1	0
37	P0317	Oriented Start Up	Yes	0	1
38	P0201	Language	English	1	1
39	P0202	Type of Control	V/f 60Hz	0	0

40	P0296	Line Rated Voltage	0-6 (Can be adjusted if overvoltage or undervoltage trips occur)	Depending on inverter model	Set as per input voltage
41	P0298	Application	Normal Duty	0	0
42	P0398	Motor Service Factor	As per motor nameplate		Motor SF
43	P0400	Motor Rated Voltage	As per motor nameplate		Motor Voltage
44	P0401	Motor Rated Current	As per motor nameplate		Motor Current
45	P0402	Motor Rated Speed	As per motor nameplate		Motor RPM
46	P0403	Motor Rated Frequency	As per motor nameplate		Motor Hz
47	P0404	Motor Rated Power	As per motor nameplate		Motor HP/KW
48	P0156	Overload Current 100% Speed	0.2 to 1.5 x Inom-ND Example: Motor FLA – 13.2A If SF is 1.0, P0156 should be set to 15.2A If SF is 1.15, P0156 should be set to 16.5A	1.05 x Inom-ND	As per motor service factor, adjust if required (SF=1.0 – 1.15 x Inom-ND) (SF=1.15 – 1.25 x Inom-ND)
51	P0159	Motor Thermal Class	Class 5,10,15,20,25,30,35,40,45	1 – Class 10	3 – Class 20

Table 2 PID SE200 Industrial Startup guide

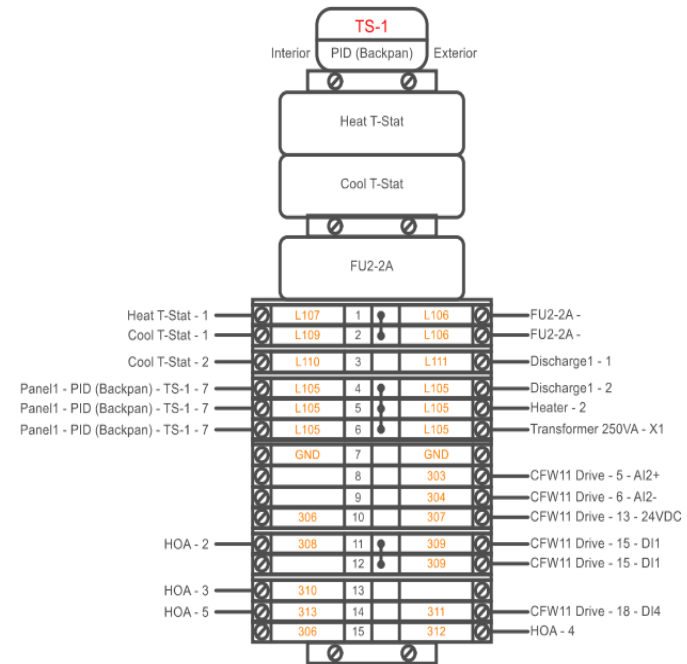
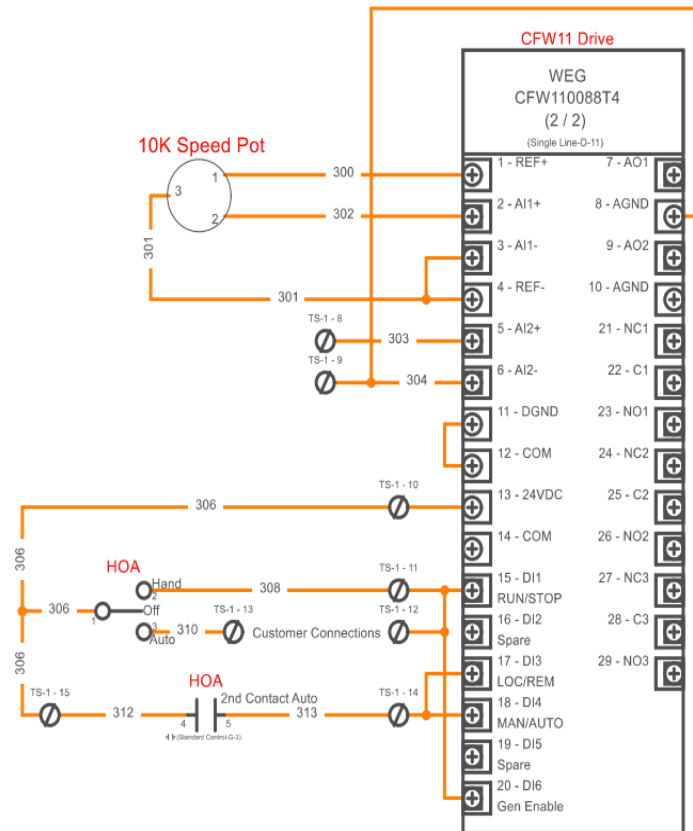


Figure 6 Industrial PID Control Schematic

Troubleshooting

For common faults/alarms and their possible causes see table starting on page 17.

Faults/Alarms	Description	Possible Causes
F006 Imbalance or Input Phase Loss	Mains voltage imbalance too high or phase missing in the input power supply. Note: - If the motor is unloaded or operating with reduced load this fault may not occur. - Fault delay is set at parameter P0357. P0357 = 0 disables the fault. - If single-phase power supply is used, this fault must be disabled.	Phase missing at the inverter's input power supply. Input voltage imbalance >5 %. For the Frame Size E: Phase loss at L3/R or L3/S may cause F021 or F185. Phase loss at L3/T will cause F006. For frame sizes F and G: Pre-charge circuit fault.
F021 DC Link Undervoltage	DC Link undervoltage condition occurred	The input voltage is too low, and the DC bus voltage dropped below the minimum permitted value (monitor the value at parameter P0004): DC BUS Voltage < 446 V - For a 440-460 V input voltage (P0296 = 3). DC BUS Voltage < 487 V - For a 480 V input voltage (P0296 = 4). Phase loss in the input power supply. Pre-charge circuit failure. Parameter P0296 was set to a value above of the power supply rated voltage

F022 DC Link Overvoltage	DC link overvoltage condition occurred.	The input voltage is too high, and the DC bus voltage surpassed the maximum permitted value: DC Bus Voltage > 400 V - For 220-230 V input models (P0296 = 0). DC Bus Voltage > 800 V - For 380-480 V input Inertia of the driven load is too high, or deceleration time is too short. Wrong settings for parameters P0151, or P0153, or P0185. Ensure P0185 and P0186 are set to 770 and 40 respectively.
F048 IGBT Overload Fault	An IGBTs overload fault occurred.	Inverter output current is too high.
F070 Overcurrent / Short-circuit	Overcurrent or short-circuit detected at the output, in the DC bus, or at the braking resistor.	Short-circuit between two motor phases. Short-circuit between the connection cables of the dynamic braking resistor. IGBT modules are shorted. Disconnect Motor try running VFD if issue persists replace inverter, if problem goes away check motor and cable for shorts perform a megger test.
F071 Output Overcurrent	The inverter output current was too high for too long.	Excessive load inertia or acceleration time too short. Settings of P0135 or P0169 and P0170 are too high
F072 Motor Overload	The motor overload protection operated. Note: It may be disabled by setting P0348 = 0 or 3	Settings of P0156, P0157 and P0158 are too low for the used motor. Motor shaft load is excessive.
F0150 Motor Overspeed	Overspeed fault. It is activated when the real speed exceeds the value of $P0134 \times (100 \% + P0132)$ for more than 20 ms.	Wrong settings of P0161 and/or P0162. Problem with the hoist-type load.

A163/164 Break Detect AI1/AI2	It indicates that the AI current (4-20 mA or 20-4 mA) reference is out of the 4 to 20 mA range.	Broken AI cable. Bad contact at the signal connection to the terminal strip. Blown control fuse on terminal strip for AI circuit. If there is no AI going to that input set the corresponding input P0233-AI1 or P0238-AI2 to 0.
A181 Invalid Clock Value	Invalid clock value alarm	It is necessary to set date and time at parameters P0194 to P0199. (Try this first if not then check battery) Keypad battery is discharged, defective, or not installed.
F185 Pre-charge Contact Fault	It indicates fault at the pre-charge contactor	Pre-charge contactor defect. Open command fuse. Phase loss in the input L1/R or L2/S. P0355 = 1 (Incorrect setting for model's frame E supplied by the DC link. For those models, the setting should be P0355 = 0.
VFD Wont Run	VFD won't run even if hand switch is on, and green button has been pressed	Verify wiring to VFD ensure 24VDC is leaving TERMINAL 13 and the circuit is completed. Ensure VFD is programmed correctly
VFD does not run off speed dial	The potentiometer isn't working to control the speed dial	Verify wiring is correct, check dip switch set point, verify programming and analog signal reference in P0018, P0019, AND P0020 For AI1, AI2, and AI3 respectively.
VFD does not run in auto when using PID mode	VFD won't run off PID only stays at min speed	Ensure there are values in P0520, P0521 and P0522, see WEG Programming manual page 20-11 for specific setpoints.

PID not working on VFD	VFD won't run off PID	Ensure the manual auto is programmed correctly, ensure that your process variable is tied into the input the PID is programmed for. Ensure that if you are in Direct/Reverse that your process is above/below the value and that if sleep mode is enabled your parameters are set correctly. Note In order to leave the sleep mode (wake-up), when in the automatic PID mode, besides the condition programmed in P0218, it is necessary that the PID error (the difference between the setpoint and the process variable) be greater than the value programmed in P0535.
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For CFW11 Alarm & Fault List refer to page 6-2 of the CFW-11 User Manual Provided

Analog Input / Output Configuration (Dipswitch's Located at top of control board)

S1.1 (Analog Output 1): Dip Switch Off - 4-20mA, Dip Switch On: 0-10vdc

S1.2 (Analog Output 2): Dip Switch Off - 4-20A, Dip Switch On: 0-10vdc

S1.3 (Analog Input 2): Dip Switch Off – 0-10vdc, Dip Switch On: 4-20mA

S1.4 (Analog Input 1): Dip Switch Off – 0-10vdc, Dip Switch On: 4-20mA

Technical Support

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Notes: